

`m4_deep_concat` (three, concat).

10. Step 8: backtest protocol (how it is trained)

Code: `04_code/src/models/deep_rolling.py` → `rolling_origin_deep()`. **Aligned verbatim with the flat baseline** to keep it fair.

10.1 Rolling-origin (walk-forward)

- **Expanding window:** `min_train=104` (first 104 weeks train-only, not tested), then test week-by-week forward;
- **Refit every 13 weeks** (`retrain_every=13`): each fold trains only on samples **before** that test week (`slice(0, i)`), strictly no future;
- each test week is predicted by its own fold model, and price reconstructed $\hat{P} = P_t \cdot \exp(\hat{r})$.

10.2 Single-fold training details (`_train_fold`)

- within the training slice, the **last 52 weeks are held out as inner-validation** for early stopping (`patience=12`);
- optimiser **Adam**, `lr=1e-3`, `weight_decay=1e-4`, `batch=32`, `epochs=80` (cap; early stopping usually triggers first);
- loss **MSE** (on standardised `r`); gradient clipping `max_norm=5.0`;
- target per-fold standardised; predictions reconstructed via $\hat{r} \cdot r_{std} + r_{mean}$.

10.3 Key hyperparameters (main model)

`lookback=4`, `d=32`, `gat_layers=2`, `tcn_layers=2`, `dropout=0.1`, `seed=42`, `fusion=gated`.

⚠ Engineering caveat: on macOS, xgboost and torch in the same process segfault due to duplicate OpenMP. So the deep scripts **read** the M1_Flat predictions (`baseline_predictions.csv`) rather than re-running xgb in-process.

11. Step 9: main results

Source: `04_code/scripts/deep/run_deep_baseline.py` → `05_outputs/baselines/Deep/_cross/deep_metrics.csv` (257 common test weeks 2021–2025, `seed=42`, `lookback=4`).

Model	RMSE	skill vs M0	DirAcc	CW vs M0	DM vs M1
M0 random walk	4.152	0.0%	–	–	–
M1_Flat_Ridge / XGB (flat ref)	4.256 / 4.368	–2.5% / –5.2%	0.498 / 0.553	0.530 / 0.104	–
M1_Deep (z_fin TCN)	4.250	–2.4%	0.494	0.315	–
M_rs_deep (z_rs Prithvi)	4.247	–2.3%	0.459	0.928	0.457
M_ship_GNN (z_ship graph)	4.168	–0.4%	0.506	0.496	0.106
M3_Deep_gated (finance+shipping, gated)	4.147	+0.11%	0.529	0.166	0.061
M2_Deep_gated (finance+RS)	4.253	–2.4%	0.475	0.769	0.485
M4_Deep_gated / M4rep (three, gated)	4.205	–1.3%	0.502	0.894	0.239
M4_Deep_Concat (three, concat)	4.320	–4.1%	0.494	0.637	0.650

RMSE = root mean squared error on price (lower is better); skill = improvement over the random walk (>0 means beating RW).

11.1 Fusion matrix (RQ2 fusion ladder: 3 modality combos × 3 fusion mechanisms)

Source: `run_deep_fusion_matrix.py` → `deep_fusion_matrix.csv` / `deep_fusion_matrix.png` (257 weeks, `seed=42`, `lookback=4`, `epochs=80`, all 9 cells run once under the same protocol). This fills the full "each fusion model × three fusion mechanisms" matrix (the three mechanisms the project implements: Encoder-Concat / Gated / Cross-Attention).

skill vs M0 (%), >0 beats the random walk:

Modality combo	Encoder-Concat	Gated	Cross-Attention
M3_Deep_gated (finance+shipping)	+0.06	+0.11	+0.74

Modality combo	Encoder-Concat	Gated	Cross-Attention
M2_Deep_gated (finance+RS)	−1.93	−2.43	−5.89
M4_Deep_gated (three modalities)	−4.06	−1.28	+0.12

Full per-cell metrics:

Combo	Fusion	RMSE	skill vs M0	DirAcc	CW vs M0	DM vs M1
M3_Deep_gated	Concat	4.149	+0.06%	0.525	0.195	0.043 ✓
M3_Deep_gated	Gated	4.147	+0.11%	0.529	0.166	0.061
M3_Deep_gated	Cross-Attn	4.121	+0.74%	0.549	0.041 ✓	0.055
M2_Deep_gated	Concat	4.232	−1.93%	0.533	0.971	0.373
M2_Deep_gated	Gated	4.253	−2.4%	0.475	0.769	0.485
M2_Deep_gated	Cross-Attn	4.396	−5.89%	0.455	0.898	0.913
M4_Deep_gated	Concat	4.320	−4.1%	0.494	0.637	0.650
M4_Deep_gated	Gated	4.205	−1.3%	0.502	0.894	0.239
M4_Deep_gated	Cross-Attn	4.147	+0.12%	0.564	0.018 ✓	0.088

New findings from the matrix:

- Finance+shipping (M3_Deep_gated) is skill>0 under all three fusions** — the strongest modality combo; its **Cross-Attention is the overall best** (+0.74%, RMSE 4.121, CWvsM0 **0.041**, significantly beating the random walk) — even better than the three-modality M4_Deep_gated, showing that adding RS is a net drag.
- Finance+RS (M2_Deep_gated) is skill<0 under all three fusions**, with Cross-Attention the worst (−5.89%) — RS noise is amplified by cross-attention, again confirming the RS modality is intrinsically weak.
- Cross-Attention only helps combos that contain shipping** (M3_Deep_gated +0.74, M4_Deep_gated +0.12, both significant CWvsM0); on finance+RS it is a disaster — the value of the fusion mechanism depends on whether the modality carries signal, not on the mechanism itself.
- Gated vs Concat**: gating clearly beats concat for three modalities (−1.28% vs −4.06%); for two modalities they are close (both carried by the shipping signal).
- ⚠ These are **single-seed (42)** results; Cross-Attention especially must be read with the multi-seed robustness — §13 shows xattn has large multi-seed variance (± 2.76 , seed2 collapses). So the main model stays **Gated**, with Cross-Attention listed as a "high-ceiling but unstable" advanced result.

12. Step 10: how to analyse the results (statistical tests)

A tiny RMSE difference means nothing without **significance testing** (04_code/src/backtest/metrics.py). The project is meticulous about **using the correct test**:

12.1 Three metrics

- RMSE skill vs M0**: $1 - \text{RMSE_model} / \text{RMSE_M0}$, >0 means more accurate than the random walk.
- Directional accuracy (DirAcc)**: share of correct up/down direction (auxiliary; not in the loss).
- RMSE, MAE**: absolute price error.

12.2 Two tests, don't mix them up (the project's rigour point)

- Clark–West (CW, for nested)**: when the small model is **nested** in the large one (e.g. "finance" nested in "finance+shipping"; the random walk $r=0$ is nested in any model). CW corrects the bias DM has for nested models.
 - CW vs M0**: honestly answers "does it beat the random walk?" — M0 is nested in any model, so it is valid.
 - Deep internal modality increments** (fusion vs fin, m4rep vs fusion, finrs vs fin) are also nested → use CW.
- Diebold–Mariano (DM, for non-nested, HLN small-sample corrected)**: when two models are **not nested** (different model class + feature set, e.g. deep vs M1_Flat, gated vs concat). Here CW is inflated; DM is correct.

Protocol revision on 2026-07-07 (important): previously "deep vs M1_Flat" was treated as nested with CW (optimistic); switching to a strict **DM(non-nested)** **downgrades** the RQ2 claim "representation-level significantly beats flat" from a strong

result to "directionally consistent but not significant". This is honest science.

12.3 How to read the results (`deep_cw.csv`)

- **Shipping increment (M3_Deep_gated vs M1_Deep) nested CW = 0.00057** ✓ — adding a shipping representation on top of finance **significantly** lowers MSE; this is RQ2's **cleanest positive evidence** that representation-level fusion helps.
- **RS increment (M2_Deep_gated vs M1_Deep) CW = 0.019** ✓, but **adds nothing on top of "finance+shipping"** (M4rep vs M3_Deep_gated CW=0.78 X, squeezed out by shipping).
- **Deep vs M1_Flat (DM) all >0.05** (M_ship_GNN 0.106 is closest): no longer significant under the strict test.
- **Gated vs concat (DM=0.22)**: gating is better but not significant; concat (M4_Deep_Concat) is the worst overall (−4.1%).

12.4 Six core findings

1. Only **M3_Deep_gated (finance+shipping) turns skill positive (+0.11%)**, the only deep model with skill>0; but CWvsM0=0.166 is still not significant — **none significantly beats M0** (the project's honest headline result).
2. **The shipping representation is the hardest positive evidence** (nested CW 0.00057), further strengthened by the GAT's O-D flow prior.
3. Against M1_Flat, switching to strict DM makes it **no longer significant** (a clear downgrade from the earlier optimistic protocol).
4. **Gating > naive concat** (the architecture ablation is directionally consistent).
5. **The RS representation is intrinsically weak**: M_rs_deep −2.3%, DirAcc 0.459 (<0.5), cls even worse (−11%), adds nothing on top of fin+ship — a frozen single-modality Prithvi helps little for **weekly** oil price (consistent with flat M2).
6. All of the above reinforce RQ2: weak increments need modality-aware fusion to add up in a consistent direction.

12.5 Flat vs Deep pairing (RQ2 writing main table)

Fix the information set (modality content), change only the architecture (deep: M1→**fin**, M2→**m2_deep_gated**, M3→**m3_deep_gated**, M4→**m4_deep_gated**; flat counterpart = same information-set XGB / Ridge):

Pair	Flat XGB RMSE	Deep RMSE	skill flat	skill deep	DM vs XGB	DM vs Ridge
M1	4.368	4.250	−5.2%	−2.4%	0.097	0.466
M2	4.440	4.253	−7.0%	−2.4%	0.042 ✓	0.096
M3	4.429	4.147	−6.7%	+0.11%	0.010 ✓	0.062
M4	4.507	4.205	−8.6%	−1.3%	0.005 ✓	0.036 ✓

How to read:

- Deep significantly beats flat XGB on **M2/M3/M4 (especially with shipping)**; M1 alone does not reach 5%.
- **Do not claim "deep is always better"** — gains concentrate in multimodal + shipping settings.
- **Coexists with §12.2–12.3**: DM vs flat **M1** is the more conservative test (fusion DM=0.061, not significant); DM vs **matched-information-set flat XGB** is the Ch4 RQ2 main table. Write with the pairing table as primary, DM vs M1 as a supplement.

13. Step 11: robustness

Script: `04_code/scripts/deep/run_deep_sweep.py` → `deep_sweep_summary.csv` (lookback=4). Goal: show the main conclusion is not a single-seed / single-hyperparameter fluke.

- **Multi-seed (3 seeds each: 42/1/2)**: fusion $-0.47\% \pm 0.86$ (**smallest variance = most stable**); m4rep $-0.89\% \pm 0.60$; m4xattn $-1.83\% \pm 2.76$ (**most unstable, seed2 collapses to −4.98%**). CWvsM0 mostly 0/3, DMvsM1 fusion 1/3 — **the stable one is gating; the unstable one is cross-attention**.
- **Hyperparameter sweep (fusion, seed42)**: lb=8 d=32 best (+0.34%, DMvsM1 0.041) > lb=4 (+0.11%) > lb=12 (negative); **d=64 is always worse** (evidence that "encoder dimension must be small"). **The main model still locks lb=4** to align the flat protocol for a fair comparison.
- **RS regularisation grid (P1-5)**: meanpool all negative (best −0.90%), cls −11% — **RS weakness is intrinsic, not fixable by tuning**.
- **Main fusion regularisation grid (P1-6)**: the whole grid is skill≈0, dp=0.3 slightly better (+0.29%) — **the main model is robust to regularisation**.
- **Optional, not blocking submission**: deep leave-one-AOI / leave-one-Hormuz-node (see progress overview §3.5).

14. Step 12: interpretability (RQ3)

Scripts: `run_deep_interpret.py` (gate + node attention) + `run_deep_xattn_viz.py` (cross-attention). During walk-forward, the fusion info dict is additionally recorded per week — never peeking at the future.

14.1 Single seed=42 main figures (narrative starting point)

- **Mean modality gate (gated main model):** finance **0.44** > RS **0.348** > shipping **0.212**. The gate weights are plotted as a time stackplot overlaid with known supply / geopolitical event lines (Russia–Ukraine 2022-02, EU RU oil ban 2022-06, OPEC+ surprise cut 2023-04, Houthi Red Sea 2023-11) to check temporal alignment.
- **Shipping node attention:** most attended is the **Hormuz chokepoint** + P003/P009/P001/P005 (matching the Red-Sea-rerouting / export-terminal narrative).
- **RS site attention:** most attended P004/P008/P009/P001/P006 (export-terminal AOs).
- **Cross-attention (finance Query over 28 tokens):** → shipping **0.575** / → RS **0.425**; top tokens are RS sites but the weights are highly uniform (~0.04), i.e. attention is "flat" → echoes weak/redundant RS information.
- **A discussion point:** the two fusion mechanisms rank "RS vs shipping" **oppositely** (gating RS>shipping; xattn shipping>RS).
- **Caveat:** a high α_{shipping} does **not** mean the model is "looking at Hormuz" — spatial detail lives in the **node/site attention** layer, not in the fusion gate itself.

Outputs (seed42): `deep_gate_weekly.csv`, `deep_interpret.png`, `deep_xattn_weekly.csv`, `deep_xattn_viz.png` (mostly under `05_outputs/baselines/Deep/M4_Deep/`).

14.2 Multi-seed gate / attention stability (2026-07-16, seeds={42,1,2})

Script: `run_deep_interpret.py --seeds 42,1,2 --lookback 4`. Outputs: `deep_gate_stability.csv` / `deep_gate_corr.csv` / `deep_gate_events.csv` / `deep_gate_band_weekly.csv` / `deep_interpret_stability.png` + `deep_gate_weekly_seed{S}.csv`.

Mean modality α (across seeds): finance $\approx 0.44\text{--}0.48$ > rs $\approx 0.30\text{--}0.35$ > shipping $\approx 0.21\text{--}0.26$ (**rank order stable**).

Week-level α_{shipping} correlation across seeds: Pearson is weak and inconsistent ($42 \leftrightarrow 1 \approx 0.10$, $42 \leftrightarrow 2 \approx 0.13$, $1 \leftrightarrow 2 \approx -0.31$) — **weekly trajectories are unstable**; do not tell fine-grained event stories from a single seed.

Event-window $\Delta\alpha_{\text{shipping}}$ (post–pre, ± 8 weeks):

Event	Same sign?	Reading
Russia–Ukraine (2022-02)	 3/3 up	Only robust "shipping-gate rises" window
EU RU oil ban / OPEC+ cut	 3/3 down	Same direction but a <i>fall</i> — do not write as supply-shock uplift
Houthi Red Sea (2023-11)	× 2↑1↓	Unstable — do not lock in

Spatial attention Top-5 frequency (stable foci safe to claim in writing):

Focus	freq	Writing stance
hormuz	3/3	 Only cross-seed-stable chokepoint focus
P003 RasTanura / P009 Ulsan / P006 Ningbo	2/3	Mention as occasional companions, not mechanisms
RS: P006 Ningbo-Zhoushan, P001 Rotterdam	3/3	 Cross-seed-stable RS sites

Writing rule: claim only **cross-seed-stable foci** (Hormuz; RS Ningbo / Rotterdam). Weak weekly α_{shipping} correlation → keep event narrative to the Russia–Ukraine co-rising window only; do not lock the Red Sea window. Keep xattn interpretability in the appendix.

15. Step 13: advanced ablations

Script: `run_deep_advanced.py` → `deep_advanced_summary.csv` (lookback=4, seed=42).

arm	skill vs M0	DirAcc	CWvsM0	DMvsM1
M4concat (concat)	−4.06%	0.494	0.637	0.650
M4rep gated	−1.28%	0.502	0.894	0.239

arm	skill vs M0	DirAcc	CWvsM0	DMvsM1
M4 cross-attention (finance Query)	+0.12%	0.564	0.018	0.088
M4rep + modality dropout 0.3	−0.19%	0.498	0.316	0.096
M4-xattn + dropout 0.3	+0.62%	0.560	0.008	0.050

- **Cross-attention can be the single-seed best:** under seed42+lb4 it turns skill positive and is the **only config that significantly beats the random walk via CWvsM0** (0.018; with dropout 0.008 and DMvsM1 0.050, doubly significant); **but multi-seed it is highly unstable (±2.76) → listed as an "upper bound exists but unstable" advanced result; the main model remains gated.**
- **Sub-periods:** xattn is a stable +0.12% in both early/late; gated is −2.35% early / turns positive +0.49% late (2023–25).

16. One-line summary + reproduction commands

One line: RQ1 increments are weak (RS especially; all models' CWvsM0 non-significant, still hard to beat M0); **RQ2 partially positive** — hardest evidence is nested CW **0.00057** for adding shipping on finance, gating > concat, and under **matched information-set pairing** deep significantly beats flat XGB on M3/M4, but under strict DM vs flat M1 it is **not significant** (directionally consistent); **RQ3** gates/attention are interpretable but must follow **multi-seed stable foci** (Hormuz; RS Ningbo/Rotterdam) — do not lock the Red Sea window. Cross-attention can be single-seed best but is multi-seed unstable → advanced.

Reproduction commands:

```
# 0) Prerequisite: run M1_Flat first (deep scripts read its predictions)
python3 04_code/scripts/flat/run_baseline.py --modality M1

# 1) Upstream data (if rebuilding)
python3 03_data/processed/M2/py/build_m2_weekly.py                # RS monthly -> weekly
python3 03_data/processed/M2/py/precompute_s2_embeddings.py      # Prithvi embeddings
python3 03_data/processed/M3/py/aggregate_shipping_to_weekly.py  # shipping -> weekly (flat GFW
+4w)
python3 03_data/processed/M3/py/build_m3_graph_weekly.py         # AOI graph intermediates
python3 03_data/processed/M3/py/build_m3_graph17.py              # 17-node graph tensor (event
+2w)

# 2) Deep main results + tests
python3 04_code/scripts/deep/run_deep_baseline.py                # main results table
python3 04_code/scripts/deep/run_deep_fusion_matrix.py          # 3x3 fusion matrix
python3 04_code/scripts/deep/run_deep_sweep.py                  # robustness sweep
python3 04_code/scripts/deep/run_deep_interpret.py --seeds 42,1,2 --lookback 4 # RQ3 + multi-
seed stability
python3 04_code/scripts/deep/run_deep_xattn_viz.py               # RQ3 cross-attention
python3 04_code/scripts/deep/run_deep_advanced.py               # advanced ablations
```

17. Appendix: full variable & parameter list

A. Data specs

Item	Value
Unified window	2019-01-04 ~ 2025-12-26 (365 weeks)
Merged matrix	365 × 213 (incl. week_ending_friday); 212 data cols = 31+55+113+11+2
Common test span	257 weeks (2021–2025)
Target	$r_{t+1} = \ln(P_{t+1}/P_t)$, reconstruct $P = P_t \cdot e^{\hat{r}}$
lookback (main)	4 weeks (07-05 first run used 8; historical / hyperparameter arm only)
RS publication lag	month-end + 15 days (as-of backward)
Shipping lags	flat GFW presence +4w ; deep-graph GFW event/O-D +2w ; PortWatch +1w ; SAR +4w

B. Model / training hyperparameters

Item	Value
Representation dim d	32 (each encoder's output)
GAT	2 layers, 4 heads, d_model=64, log O-D flow prior
TCN	2 layers, kernel=3, adaptive dilation (dilation=1 at lb≤5)
RS encoder	frozen Prithvi-EO-2.0-300M (300M params) + temporal/site attention, proj 1024→64→32
Fusion	gated (main) / concat / cross-attention (4 heads)
Optimiser	Adam, lr=1e-3, weight_decay=1e-4
batch / epochs	32 / 80 (inner-val early stopping, patience=12)
Gradient clip	max_norm=5.0
Backtest	min_train=104, retrain_every=13, seed=42 (main table); robustness also seeds {1,2}

C. Graph nodes and chokepoints

- **11 AOI:** P001 Rotterdam, P002 Fujairah, P003 RasTanura, P004 Jurong, P005 Houston, P006 NingboZhoushan, P007 Jamnagar, P008 Basra, P009 Ulsan, P010 Kharg, P011 Yanbu.
- **6 chokepoints:** hormuz, suex, malacca, mandeb, panama, cape.

D. Key files

Category	Path
Encoders	04_code/src/models/{finance,rs,shipping}_encoder.py
Fusion	04_code/src/models/fusion.py
Data alignment	04_code/src/models/deep_dataset.py
Backtest kernel	04_code/src/models/deep_rolling.py
Entry scripts	04_code/scripts/deep/run_deep_{baseline,sweep,interpret,advanced,fusion_matrix,xattn_viz}.py
Graph tensor	03_data/processed/M3/outputs/m3_graph17_tensors.npz
Prithvi embeddings	03_data/processed/M2/outputs/s2_prithvi_emb_meanpool.npy
Results	05_outputs/baselines/Deep/{M*_Deep,_cross}/
RQ3 multi-seed	.../M4_Deep/deep_gate_stability.csv etc.
Progress overview	00_admin/最新待整理/2026-07-15_研究方案与进度总览.md
Flat walkthrough	00_admin/最新待整理/flat_baseline_full_walkthrough_EN.md